

BANK OF AMERICA CONSUMER COMPLAINTS ANALYSIS (2017–2023) (Data used for Mentorship Programme)

Business Intelligence Project Report

Tools Used: **Microsoft Excel** | **Power BI**

Data Source: **Consumer Financial Protection Bureau (CFPB)**

Date: July 2026

Executive Summary

Customer complaints are a critical indicator of service quality, operational efficiency, and regulatory compliance within financial institutions. Effectively analyzing complaint data enables organizations to identify recurring issues, improve customer satisfaction, and make informed business decisions.

This project analyzes **62,516 customer complaints** submitted against **Bank of America** between **2017 and 2023**, using data obtained from the **Consumer Financial Protection Bureau (CFPB)**. The analysis explores complaint distribution across states, products, and submission channels while evaluating the bank's response performance over time.

Using **Microsoft Excel** for data preparation and **Power BI** for visualization and business intelligence reporting, the project transforms raw complaint records into actionable insights that support operational improvement, customer experience enhancement, and strategic decision-making.

1. Introduction

Customer complaints provide valuable feedback on an organization's products, services, and operational processes. Rather than viewing complaints solely as service failures, organizations can leverage them as a strategic resource for identifying weaknesses and driving continuous improvement.

The objective of this project is to analyze historical customer complaint data to uncover patterns, evaluate response performance,

and provide evidence-based recommendations that can improve service delivery.

2. Business Problem

Financial institutions receive thousands of customer complaints annually, making it difficult to identify recurring issues, measure operational performance, and proactively resolve systemic problems.

Without effective analysis:

- Recurring issues remain unnoticed.
- Customer dissatisfaction increases.
- Operational inefficiencies persist.
- Regulatory risks become more difficult to manage.
- Management lacks reliable information for strategic decision-making.

This project addresses these challenges by transforming customer complaint data into meaningful business intelligence.

3. Project Objectives

The project aims to:

- Identify states generating the highest number of complaints.
- Determine products and services attracting the most complaints.
- Analyze complaint trends over time.
- Evaluate the organization's complaint response performance.
- Generate actionable recommendations to improve customer satisfaction and operational efficiency.

4. Business Questions

The analysis seeks to answer the following questions:

1. Which states generated the highest volume of customer complaints?
2. Which banking products received the highest number of complaints?
3. How have complaint volumes changed over time?
4. How effective is the bank in responding to customer complaints?
5. What opportunities exist to improve customer experience and complaint management?

5. Dataset Description

Dataset

Bank of America Consumer Complaints (2017–2023)

Source

Consumer Financial Protection Bureau (CFPB)

Dataset Overview

Metric	Value
Records	62,516
Variables	12
Period Covered	2017–2023

Variables Included

- Complaint ID
- Date Submitted
- Date Received
- State
- Product
- Sub-product
- Issue
- Sub-issue
- Submitted Via
- Company Public Response
- Company Response to Consumer
- Timely Response

6. Data Cleaning and Preparation

Before analysis, the dataset underwent extensive cleaning to improve quality and consistency.

Data Quality Issues

Missing Values

Missing values were identified in the following fields:

Field	Missing Values
Sub-product	7
Sub-issue	10,859
Company Public Response	1,921
Timely Response	1,494

State Names

All state names appeared as abbreviations and were converted to their full names to improve readability during visualization.

Duplicate Records

No duplicate records were identified.

7. Data Cleaning Process

The following activities were completed:

- Converted the raw dataset into an Excel Table.
- Checked for duplicate records.
- Standardized state names.
- Replaced blank values in **Timely Response** with **Pending** because these complaints were still open.
- Replaced missing values in **Sub-product**, **Sub-issue**, and **Company Public Response** with **Unknown** to preserve data integrity while indicating unavailable information.

8. Data Analysis

The cleaned dataset was imported into Power BI for analysis.

The following measures were created using DAX:

- Total Complaints
- Total Products
- Total States
- Timely Responses
- Untimely Responses
- Pending Responses
- Timely Response Rate
- Pending Response Rate

A dedicated Date Table was also created to support time-based analysis.

9. Dashboard Development

Two interactive dashboards were developed.

Dashboard 1 – Customer Complaints Overview

The first dashboard provides a high-level overview of complaint distribution across the organization.

Visualizations include:

- KPI Cards
- Top 10 Complaints by State
- Top 10 Products
- Monthly Complaint Trend
- Complaints by Submission Channel

- Complaints by Issue
- Interactive Filters

Dashboard 2 – Complaint Trends and Response Performance

The second dashboard focuses on operational performance.

Visualizations include:

- Response Performance KPIs
- Timely Response Distribution
- Quarterly Complaint Trend
- Timely Responses by Product
- Product Response Matrix
- Interactive Filters

10. Key Findings

1. Customer complaints are heavily concentrated in a small number of states.

California recorded **13,709 complaints**, followed by Florida (**6,488**) and Texas (**4,686**).

2. Checking and Savings Accounts generated the highest complaint volume, accounting for **24,814 complaints**.
3. The bank achieved an impressive **93.77% Timely Response Rate**, indicating strong operational responsiveness.
4. Despite strong response performance, **2,403 complaints were handled late**, while **1,494 complaints remained pending**, highlighting opportunities for further process improvement.
5. Complaint volumes declined during the fourth quarter, suggesting improvements in customer service or complaint prevention initiatives.

11. Recommendations

Based on the analysis, the following recommendations are proposed.

Service Improvement

Focus operational improvement initiatives in California, Florida, and Texas, where complaint volumes are highest.

Product Enhancement

Conduct a detailed review of Checking and Savings Account processes to eliminate operational bottlenecks contributing to customer dissatisfaction.

Root Cause Analysis

Move beyond reactive complaint resolution by identifying and eliminating the underlying causes of recurring issues.

Response Performance

Implement Service Level Agreement (SLA) monitoring and prioritize clearing pending complaints to further improve customer satisfaction.

Continuous Improvement

Study the practices of low-complaint regions and replicate successful operational strategies across higher-performing branches.

12. Business Value

This analysis enables management to:

- Improve customer satisfaction
- Reduce complaint volumes
- Strengthen operational efficiency
- Enhance regulatory compliance
- Support evidence-based decision-making
- Improve service quality through proactive intervention

13. Conclusion

This project demonstrates how business intelligence can transform large volumes of customer complaint data into actionable insights.

By combining **Excel** and **Power BI**, the analysis provides a comprehensive view of customer complaints, operational performance, and service quality. The findings highlight both strengths, such as the bank's high timely response rate, and opportunities for improvement, particularly in product optimization, regional service delivery, and proactive complaint prevention.

Implementing the recommendations presented in this report will help reduce complaint volumes, enhance customer experience, strengthen operational efficiency, and support continuous improvement across the organization.

Appendix

Tools Used

- Microsoft Excel
- Power BI

Data Source

Consumer Financial Protection Bureau (CFPB)

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